



DAFFODIL INTERNATIONAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FYDP (PHASE-I) EVALUATION REPORT

REPORTING PERIOD -SPRING 2025

Project Identification:

I. Project Title	Book Shop - A Online E-Commerce Platform for Buying and Selling Books	
II. Group Members	Name: S.M. Hasan Ul Islam	Student ID: 213-15-4599
III. Supervisor	Name: Ms. Hasnur Jahan Designation: Lecturer	
IV. Co-Supervisor	Name: Ms. Tasfia Anika Bushra Designation: Lecturer	
V. Submission Date:	13 th January, 2026	
VI. Certificate:	“This is to certify that the final year design project work until Phase-I evaluation held on 13-01-2026, titled as stated in Sec. I, executed by the students’ group mentioned in Sec. II, have been found satisfactory and every section of this report is reflecting the same.”	<i>Hasnur</i> 13.1.2026 (Signature of Supervisor & date)

Project Insights

Thematic Area(s): <i>[Just click the check box]</i>	Artificial Intelligence and Machine Learning	☐
	Deep Learning	☐
	Health Informatics	☐
	Cybersecurity	☐
	Software Engineering and Development	☒
	Blockchain Technology	☐
	Internet of Things (IoT)	☐
	Computer Networks	☐
	Computer Vision	☐
	Natural Language Processing (NLP)	☐
	Robotics	☐
	Game Development	☐
	Cloud Computing	☐
	Image Processing	☐



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	Robotics	☐
	Game Development	☐
	Cloud Computing	☐
	Image Processing	☐

	Others (please specify):
Software packages, tools, and programming languages	Programming Languages: - TypeScript 5.x, - JavaScript (ES6+), - HTML5, - CSS3 Frontend: - Next.js 15.5.9, - React 19.0.0, - Tailwind CSS 4.1.7, - Redux Toolkit 2.8.2, - React Hook Form 7.62.0, Backend: - Node.js, - Express.js 5.1.0, - MongoDB, - Mongoose 8.18.1, - Passport.js 0.7.0, JWT, - Multer 2.0.2, - Cloudinary 2.7.0, - SSLCommerz LTS 1.2.0, Development Tools: - npm, - Git, - Visual Studio Code, - Nodemon 3.1.10, - ts-node 10.9.2 Third-Party Services: - Cloudinary (Image Storage), - SSLCommerz (Payment Gateway), - Google OAuth (Authentication), - Gmail SMTP (Email Service) Testing & Documentation: - Postman / Insomnia, - Markdown

CO Description for FYDP-Phase-I

CO	CO Descriptions	PO
CO4	Perform economic evaluation, cost estimation, and apply suitable project management procedures throughout the FYDP lifecycle in the context of developing the multi-vendor Book-Shop website.	PO11
CO6	Select and apply appropriate methodologies, resources, and contemporary engineering/IT tools for prediction, modeling, and solving complex engineering processes for the Book-Shop (multi-vendor) project.	PO5
CO7	Assess societal, health, safety, legal, and cultural issues and responsibilities in professional engineering practice related to the FYDP problem.	PO6
CO10	Operate effectively as an individual and as a member/leader in multidisciplinary teams during FYDP.	PO9

1. Project Overview

1.1 Introduction

Bangladesh created a full-stack website called Book Shop to make buying and selling books faster. The platform's easy-to-use design makes it easy for students, book readers, and sellers to trade both new and used books. The system of this book-shop will allow user to see the book-list for sale, browse the inventory, manage wishlists and shopping carts, process safe payments, and manage orders.

1.2 Background Study

The inspiration come from the observation of DIU fair which is held every Monday in a week at the AB4building, There are many kinds stall but a few were standing in front of book stall. It is that moment I have to create a website which is depend on one and only pure book-sop, no other things will be mixed, The academic thing s will also be seen in website, but for now only books will be added. Making it possible, there are several critical challenges and opportunities were identified:

Challenges Observed:

- **Accessibility issue:** Many students were unable to participate in a book fair because of time and location or any other obstacles.
- **Inventory Issues:** Sellers could not found the available books, because of there is no listing or any organized system. manage inventory and that result is late responds.
- **Price bargaining issue:** Those discussions are held in-person, took a lot of time and lost the chances for both buyers and sellers.
- **Payment limited issue:** Payment choices were restricted and security issues were raised by cash-only transactions.
- **Less communication issue:** There is no way, that any other students who miss the chance to buy or sell books after the fair finished.
- **Information issue:** All buyers had difficulty finding specific books, while sellers couldn't effectively reach their target audience.

Opportunities Identified:

- **Demanding:** The student has a strong will to get a book to read and after finishing it , they sell that book either if that is academic and any regional books.
- **Active Community:** DIU has a fabulous admitted student campaign that could represents a significant potential user base.
- **Digital Readbook:** Though every students who is well known by this modern life, specially the Gen-Z, they are addicted to mobile phones and online things. It is high possibility to make many customer in virtual.
- **Cost-Effective Solution:** A sustainable market for a book exchange was shown by students' preference for reasonably priced used books.

According to the growth of e-commerce in Bangladesh, the digital marketing has expanded rapidly, with online transactions that has been increase by more than 40% annually. Digital platforms are essential for business because the COVID-19 pandemic accelerated the shift to online shopping. This trend supports the potential of an online book trading platform in Bangladesh.

Second-Hand Book Market Research:

In this project, I ran a research on old books buying and selling ratio, the result absolutely shocking to me. The old books that could be any types of books that are more valuable than a new ones. Here is some studies that I have gained from my research:

- The number is Greater than 60% who suggest every reader or students should buy old textbooks at a lower price.
- Most of the students sell books after completing courses or manage a specific time to sell books, and make the supply continue running.
- The awareness among younger generations are mostly use the old ones.
- The comparison between an fair costs by 30% to 40% in digital marketing.

Online Marketplace Best Practices:

The key factors is highlighting the literature on this book related e-commerce platforms:

- **User Trust:** Every customer want their privacy in terms of like payment and user authentication and 2FA verification.
- **Search and Discovery:** The user experience (UX) and transaction rates have improved the functionality.
- **Mobile-First Design:** Firstly, we develop website with responsive design which is support the ratio of any mobile, so it would be more interactive with youth.
- **Payment Integration:** At this moment the government open a way to payment easily in globally via PayPal. It would be a great opportunity for us like influencers, freelancers. In 2026, the Paypal, GPay will be new payment method added beside bKash, and bank system.

Academic E-Commerce Platforms:

Studies on university-based e-commerce platforms show that,

- Those marketplaces have students, found that it'd have better trust and transaction success rates than any places.
- Academical platforms have 25-35% higher user rates compared to general marketplaces.
- The availability and engagement have been improved with improvement of academic communities.
- Having a better availability and participation when works closely with universities and researchers.

1.3 Gap Analysis

As per my research for this project, there are few book store in Bangladesh, where anyone can buy book, but could not sell the book what they have finished, or any other reasons. Based on this background study, the following gaps exist between current solutions:

Gap ID	Problem Area	Current State (Manual/Informal)	Proposed Target (Book-Shop)	Software Implementation (Features)
G1	Academic Focus	General e-commerce lacks focus on specialized academic books.	Dedicated platform for academic materials and community exchange.	Category-based navigation; Peer-to-Peer (P2P) listings.
G2	Time & Geo Limits	Book fairs (like Feb 21st) are seasonal and location-bound.	24/7 access from any location in Bangladesh.	Cloud-hosting; Next.js SSR for high availability and performance.

G3	Search & Discovery	Social media/fairs lack structured search and filters.	Advanced multi-criteria search for specific versions.	Full-text search engine; Filters for Author, Edition, and Condition.
G4	Payment Security	Cash-only or unsecured transfers in informal groups.	Secure, consolidated infrastructure with multiple options.	SSLCommerz Integration (bKash/Nagad); Escrow logic.
G5	Trust & Verification	No user history or validation in informal environments.	Trust-based ecosystem with verified reputations.	User Profiles & Rating Systems; Verified Seller Badges.
G6	Continuity	Market activity dies after physical fairs conclude.	Persistent marketplace for year-round trading.	MongoDB Persistent Inventory; Automated listing management.
G7	Mobile Optimization	Existing systems are desktop-heavy and slow on local data.	Mobile-first design for 80%+ mobile usage in BD.	Responsive CSS/Tailwind; Asset optimization for low-bandwidth.
G8	Cost-Effectiveness	Traditional shops add high markups and fixed taxes.	Direct P2P transactions to eliminate middleman fees.	Real-time Chat; Dynamic pricing labels and student discount toggles.

In my perspective, this gap analysis is not explains all, but it does highlight the importance of a specialized online book selling buying platform that overcome these restrictions and provides a full answer for the academic community and readers.

2. Objectives

I want to develop what actually are the goals to get something for this project. I will write that in points for your better understanding:

Primary Objectives:

- Create a E-Commerce transactions solution Blog or Book.
- Enable the user account protection with email verification and password reset.
- Implement on the payment Gateway(SSLCommerz), for secure the financial transactions.
- Develop a User-Friendly Interface
- Implement Shopping Cart and Wishlist Functionality
- Build an Order Management System with status tracking
- Facilitate Seller Product Listings and Management

Secondary Objectives:

- Use Google OAuth for a fast login
- Support various payment options: SSLCommerz, bKash, Nagad, PayPal, GPay etc.
- Add image uploading system
- Implement a responsive design for both OS mobile and desktop.
- Sent email tips and updates for order status with verification.
- Will create admin dashboard for platform management.

3. Methodology/ Requirement Specification:

3.1 Research Design/ Prototype Design

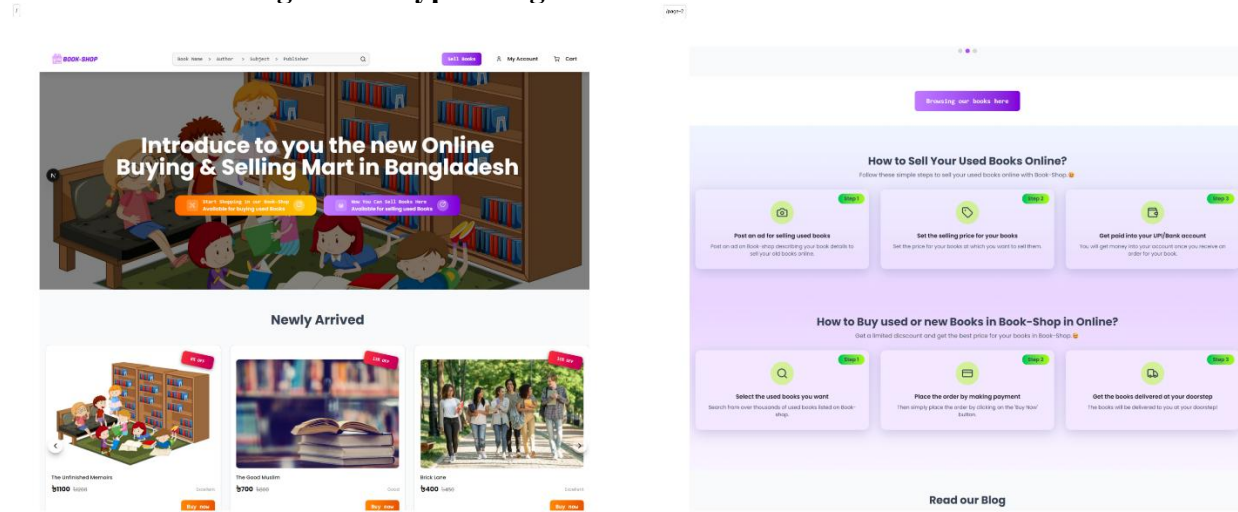


Figure: 1.0.1 – Figma Design Layout-1

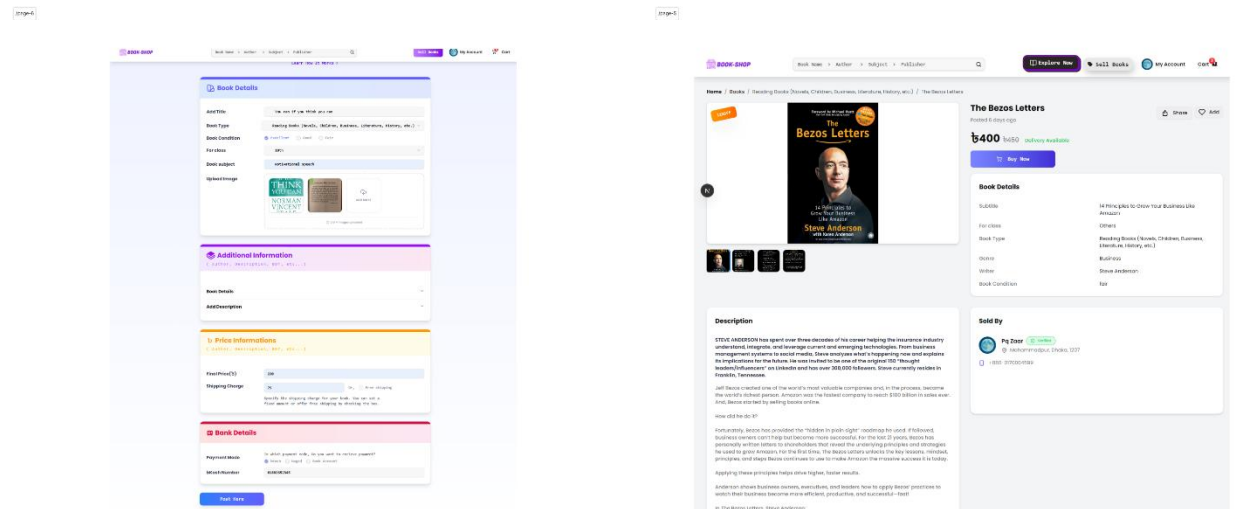


Figure: 1.0.2 – Figma Design Layout-2

Image-3

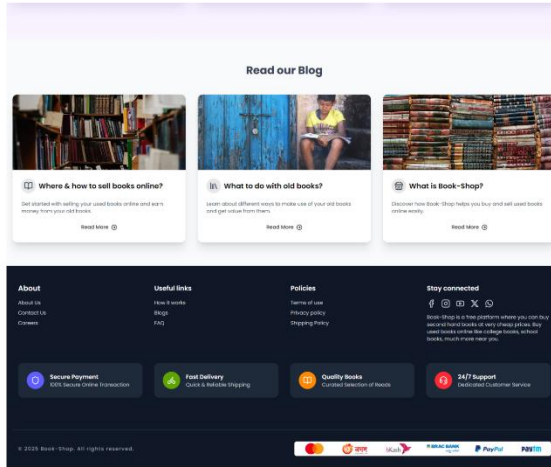


Image-4

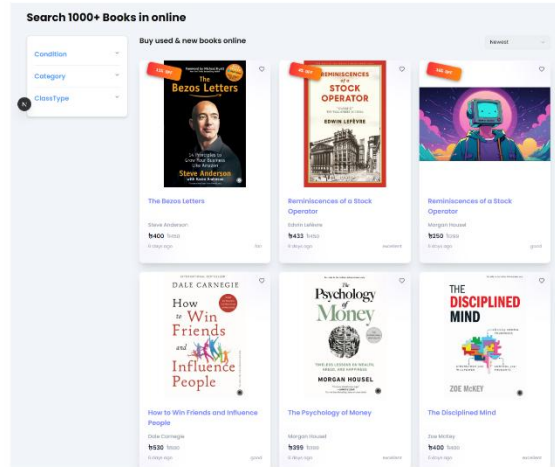


Figure: 1.0.3 – Figma Design Layout-3

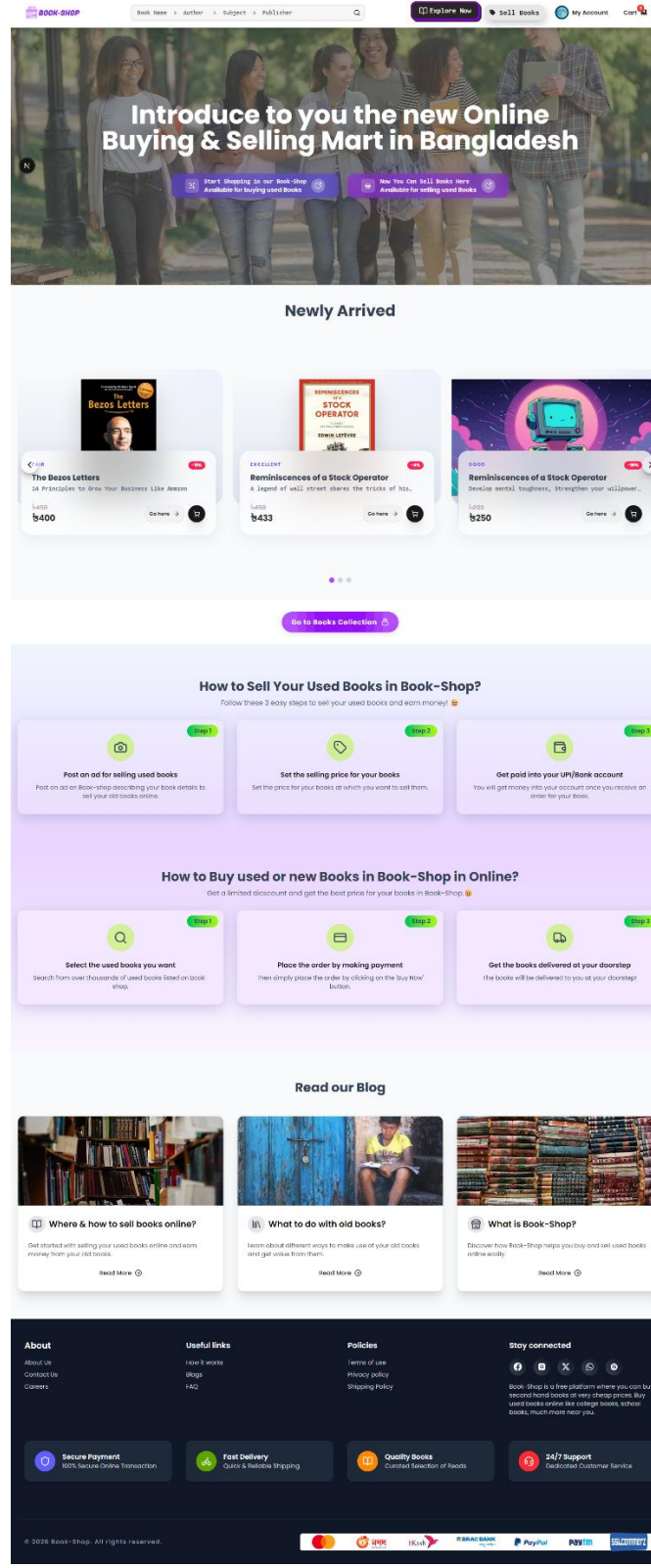


Figure: 1.1.1 – Homepage Design Layout

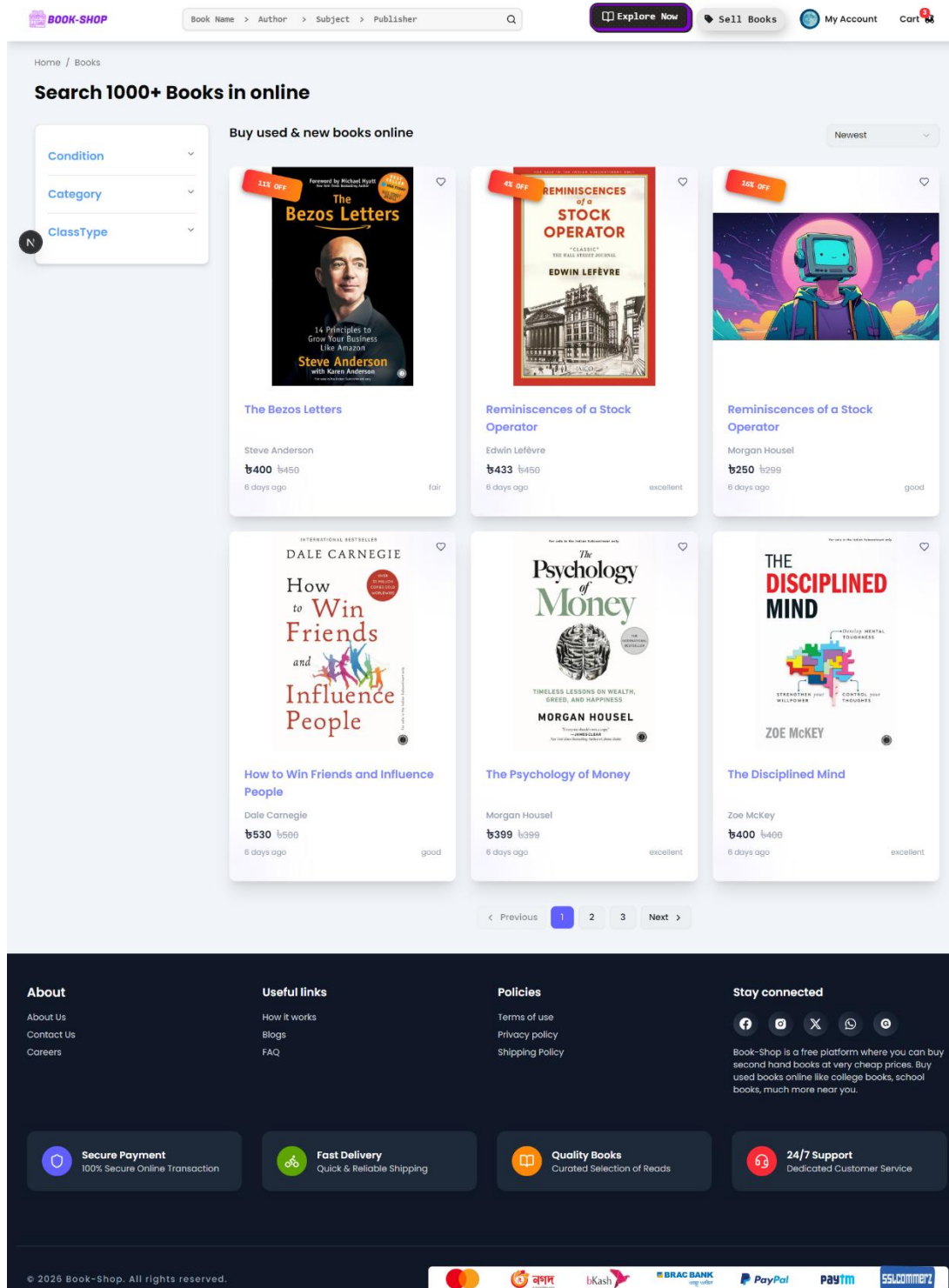


Figure: 1.1.2 – Books Design Layout

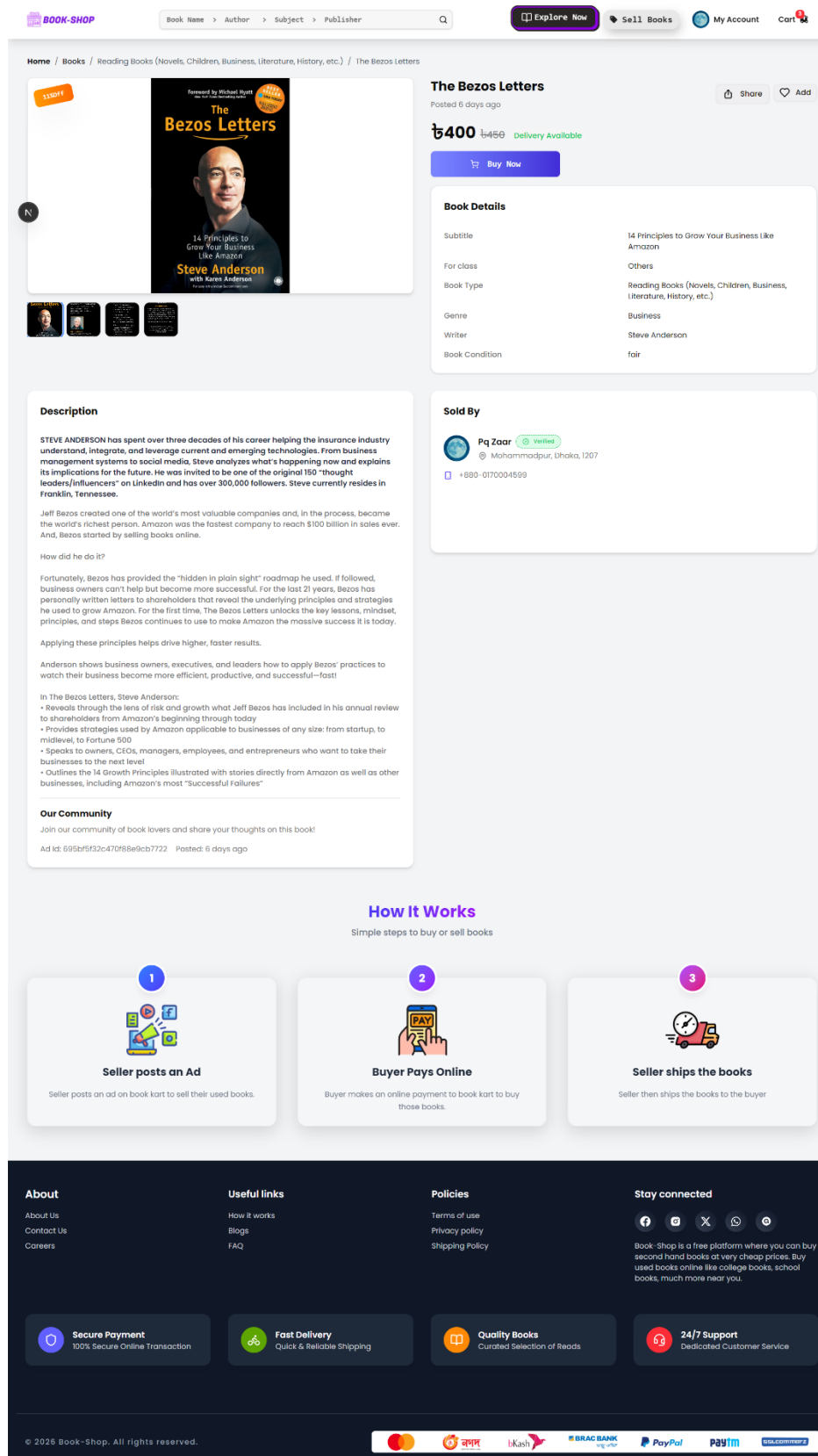


Figure: 1.1.3 – Selected Book Design Layout

BOOK-SHOP

Book Name > Author > Subject > Publisher

Q

Sell Books

My Account

Cart

Learn How It Works >

Book Details

Add Title

You can if you think you can

Book Type

Reading Books (Novels, Children, Business, Literature, History, etc.)

Book Condition

☒ Excellent
 ☐ Good
 ☐ Fair

For class

10th

Book subject

Motivational speech

Upload Image

ADD MORE

2 of 4 images uploaded

Additional Information

(Author, Description, BDT, etc....)

Book Details

Add Description

Price Informations

(Author, Description, BDT, etc....)

Final Price(b)

220

Shipping Charge

75

Or, ☐ Free shipping

Specify the shipping charge for your book. You can set a fixed amount or offer free shipping by checking the box.

Bank Details

Payment Mode

In which payment mode, do you want to receive payment?

☒ bKash
 ☐ Nagad
 ☐ Bank Account

bKash Number

01882652345

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Book-Shop is a free platform where you can buy second hand books at very cheap prices. Buy used books online like college books, school books, much more near you.

Secure Payment

100% Secure Online Transaction

Fast Delivery

Quick & Reliable Shipping

Quality Books

Curated Selection of Reads

24/7 Support

Dedicated Customer Service

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Figure: 1.1.4 – Homepage Design Layout

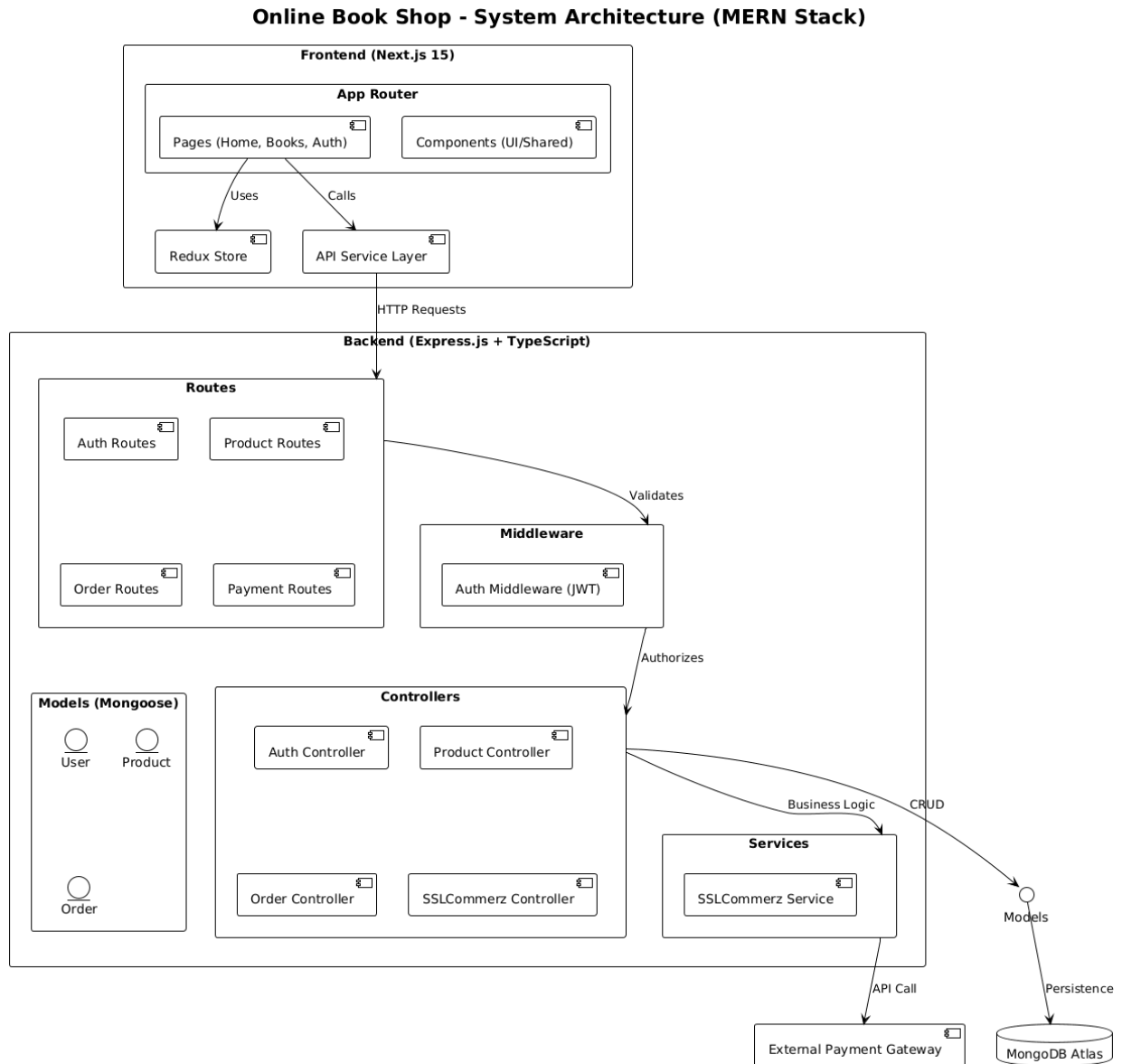
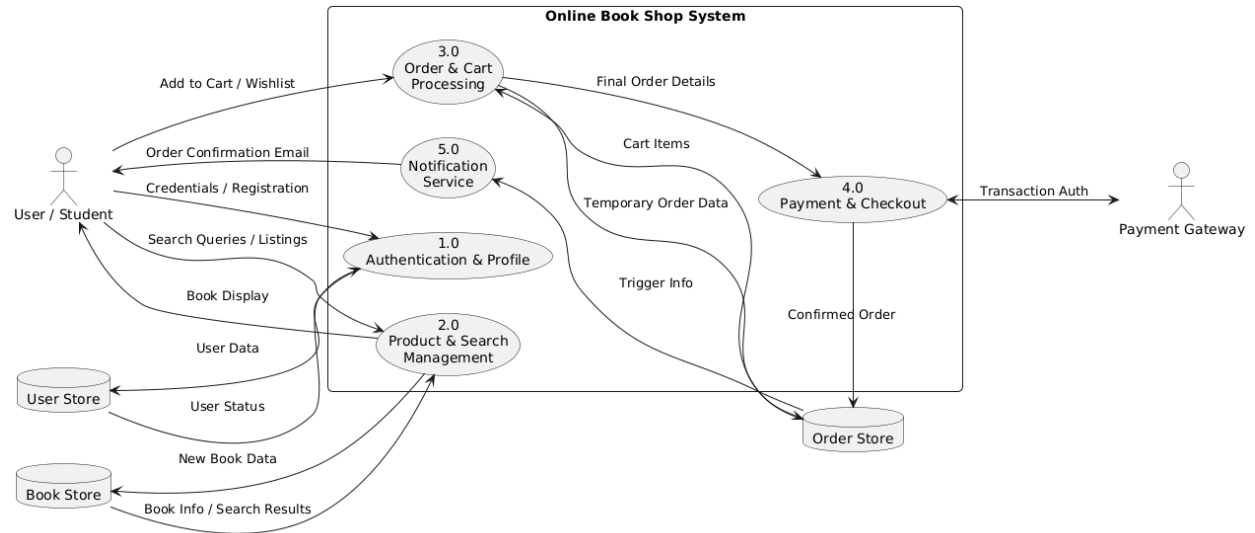


Figure: 1.2 – System Architecture Diagram

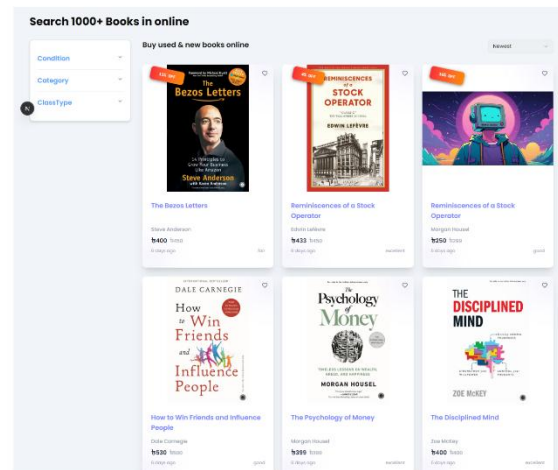
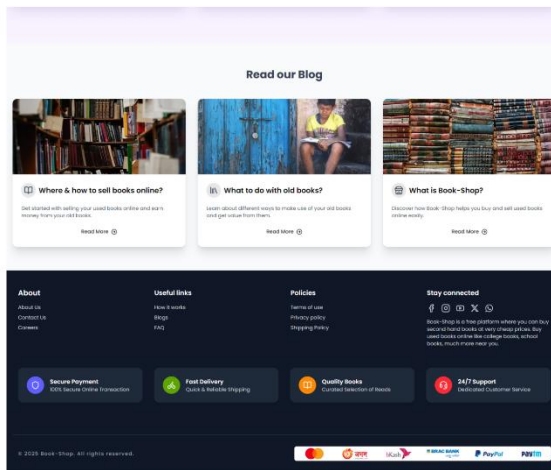
UML Designs:

Data Flow Diagram:



[page 3]

[page 4]



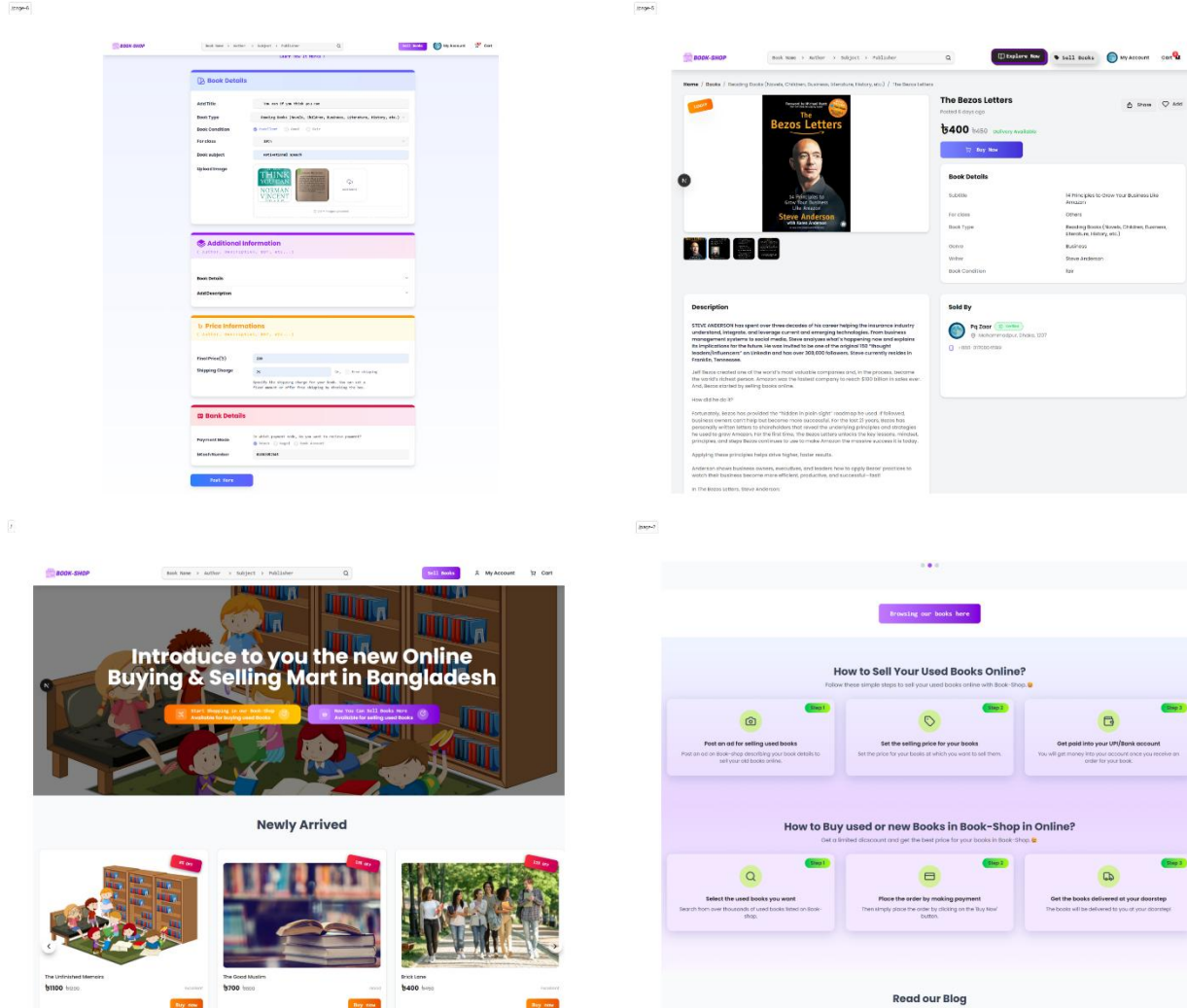


Figure: 1.3 – Data Flow Diagram

Use Case Diagram:

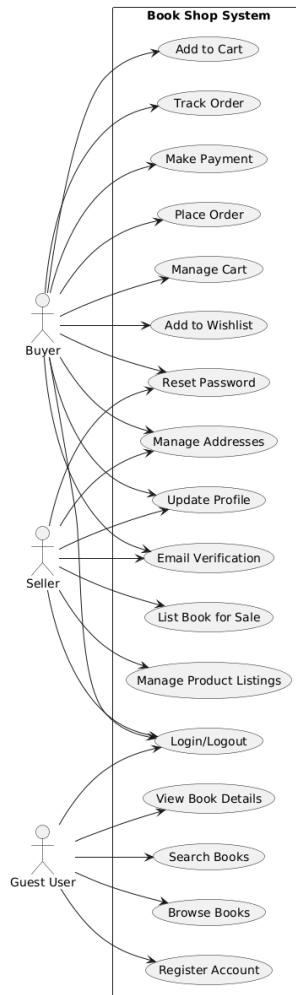


Figure: 1.4 – Use Case Diagram

Activity Diagram - Order Processing:

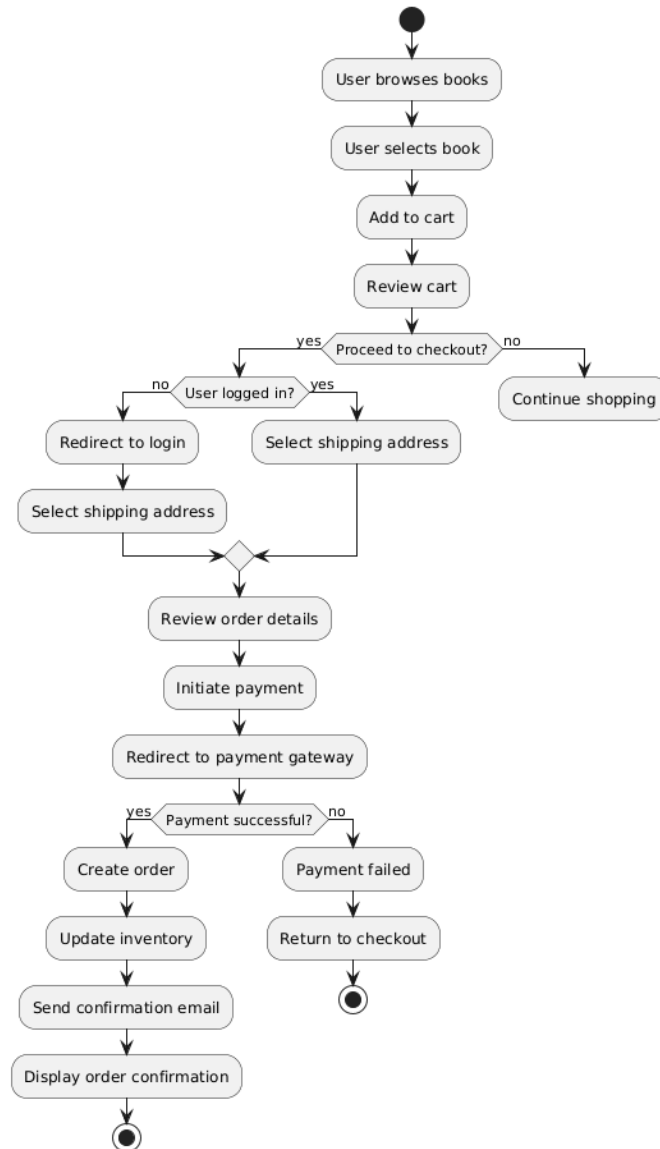


Figure: 1.5 - Activity Diagram (Order Processing)

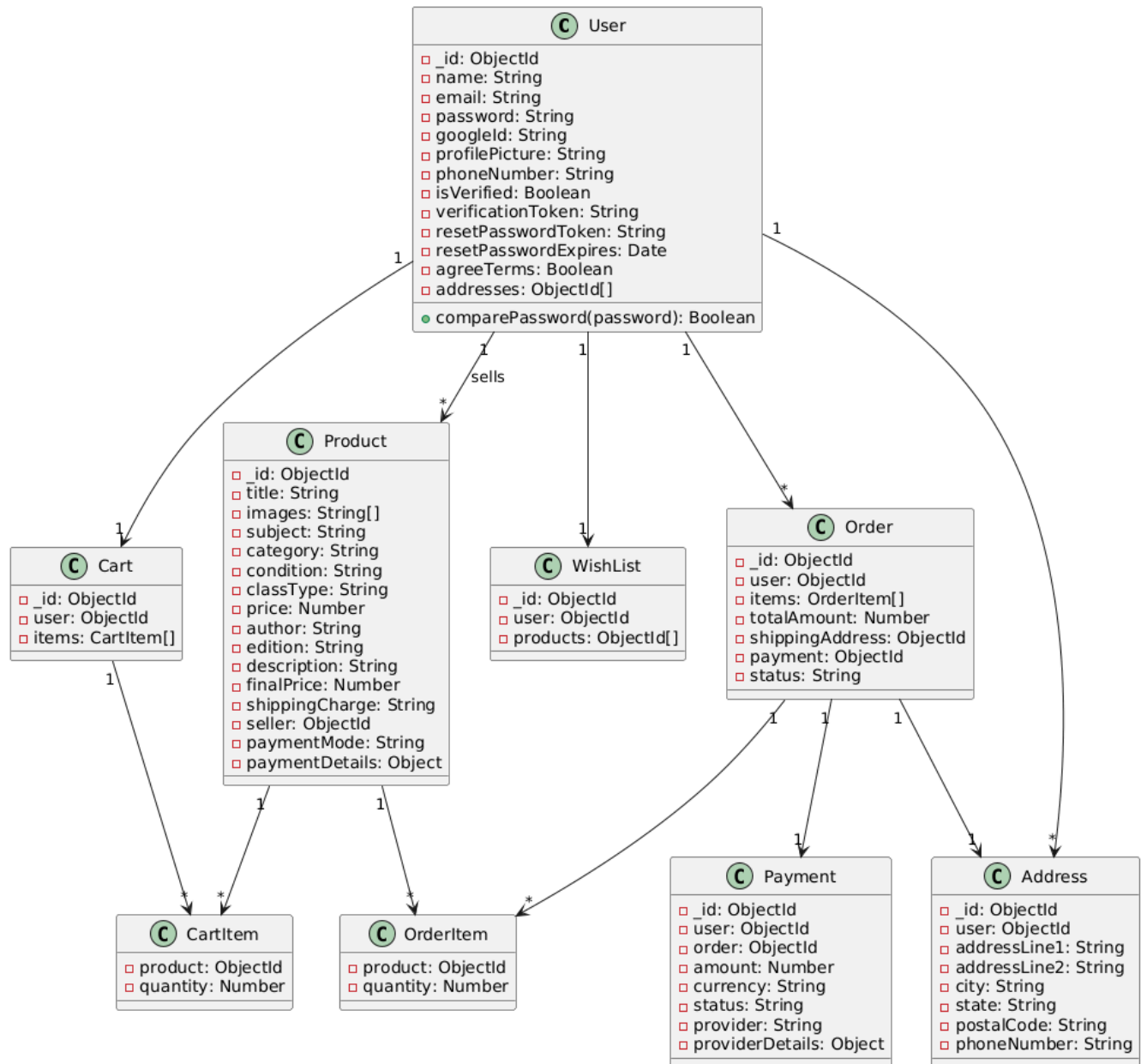
Class Diagram:

Figure: 1.6 - Class Diagram

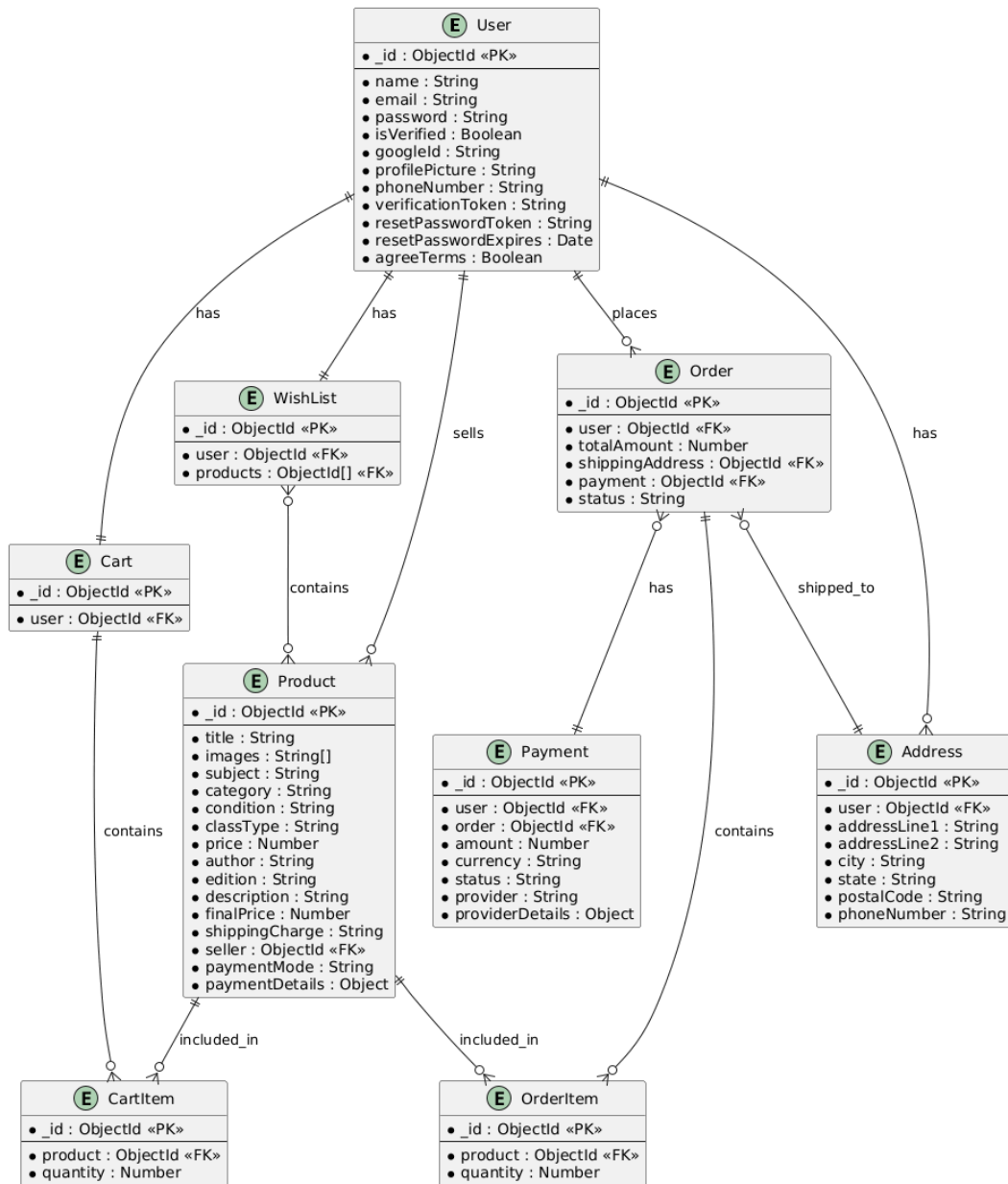
ER Diagram:

Figure: 1.7 – ER Diagram

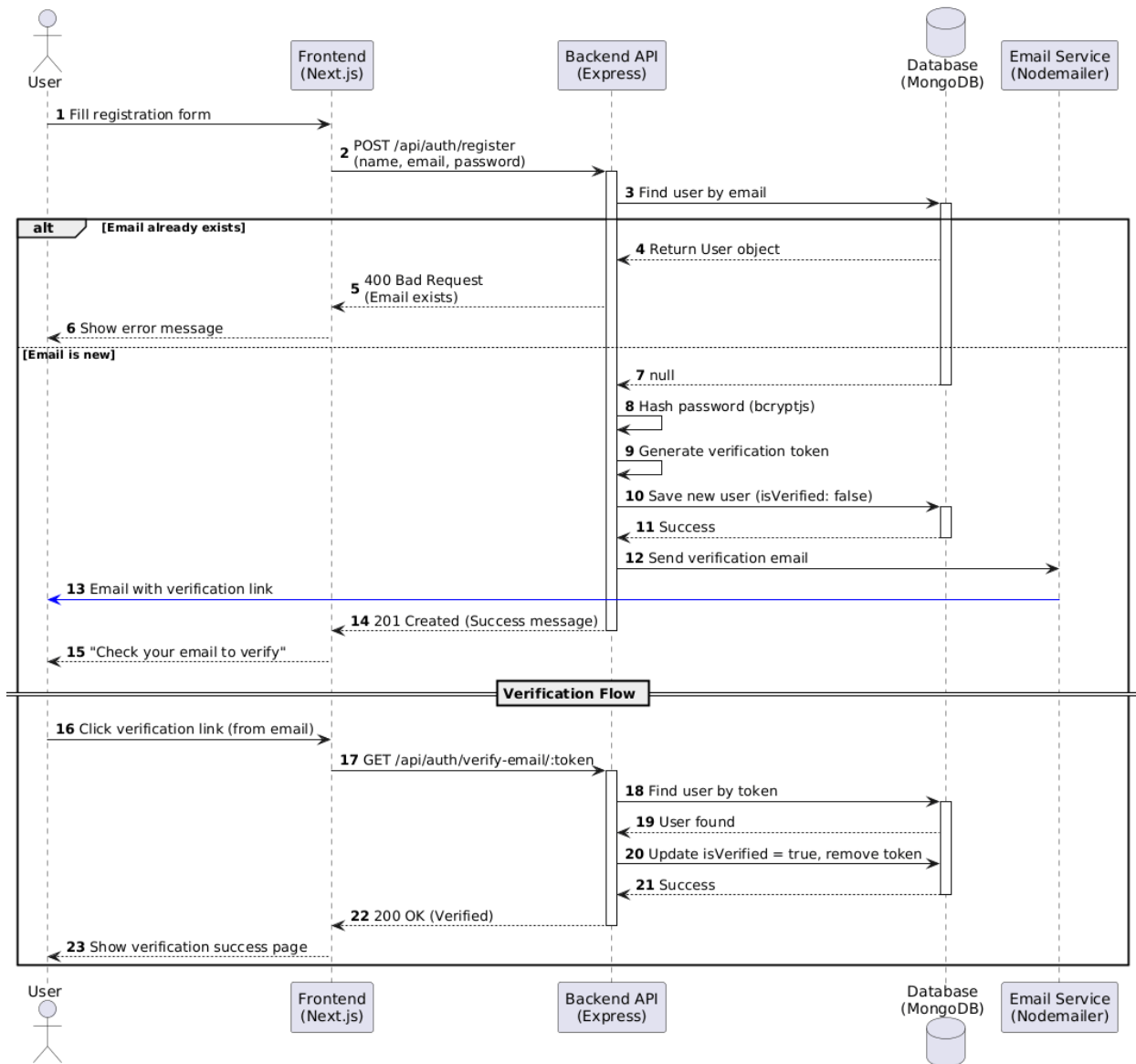
Sequence Diagram(user registration and email verification):

Figure: 1.8 - Sequence Diagram (user registration and email verification)

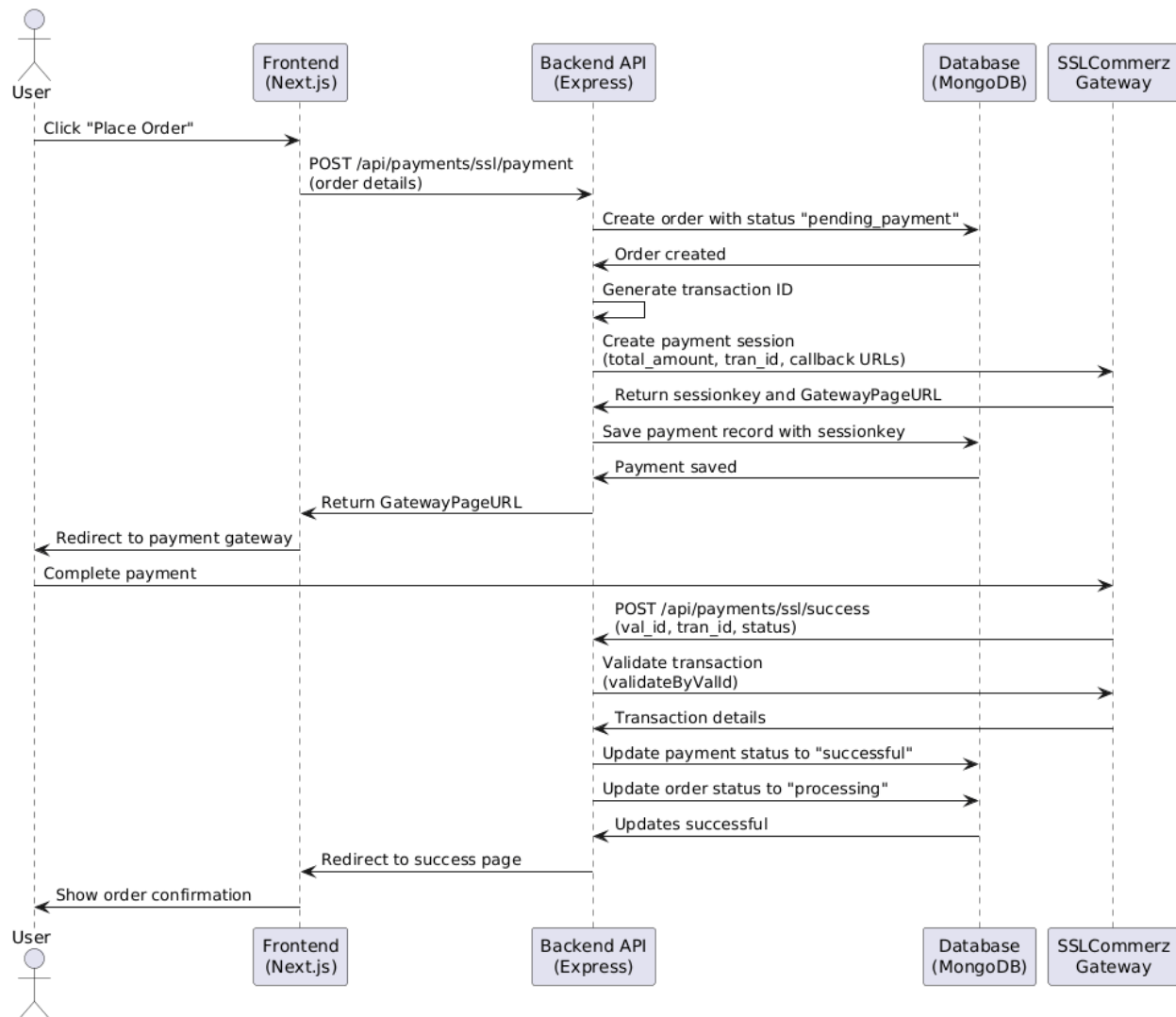
Sequence Diagram(payment processing):

Figure: 1.9 - Sequence Diagram (Payment Processing)

3.2 Data Collection

The process for collecting requirements used a complex plan for objectives that included all stakeholders in the requirements gathering process as well. This complex plan included comparing Amazon, Book-Finder, and Local Marketplaces to identify the UI/UX standards and gaps that exist (an example of this is the lack of filtering for Academic Editions versus higher delivery charges for students). In addition to students buying and selling books, faculty conducting research and regular readers browsing, profiles were developed for "imagined" users looking for things such as low prices for used textbooks. Besides, nowadays modern stacks such as MERN or the Next.js were tested for security, focusing on SSL encryption, JWT-based authentication, and secure payment methods.

Summary of Tools & Techniques:

Tool/Technique	Purpose
Surveys/Questionnaires	Gathering statistics on consumer buying and price elasticity.
User Flow Mapping	Imagined how we were going to get that book and buy it.
SWOT Analysis	SWOT analysis of the systems if there are strengths, weaknesses, opportunities and threats
Prototyping	In the early stages, the potential user is selected after validating the functional requirements with the wireframes.

➤ Final Requirement Specification

The essential requirements were formalized to guide the development process which is based on the assessment:

1. Functional Requirements

- **User Management:** Creating a user account, the user should use email verification and set password also for enhancing the account security.
- **Commerce Engine(essential functionalities for running):** An adaptable shopping cart, wish list, and multiple-step purchase procedure.
- **Seller Portal:** In this part, there will be book exposure, database management along with the image uploads.
- **Logistics & Payments:** It will have a well organized address management process, order tracking, and secure payment processing.

2. Non-Functional Requirements

- **Security:** Adding a higher and strong authentication implements and data protection for payment credentials key.
- **Performance:** Websites have it own API, so it takes time to render images and optimized responsiveness for all OS devices. (Mobile/Tablet/Desktop).
- **Dependability:** There is alert system in this store, so that any user could have been alert for any security breach or transaction.

3.3 Analysis Techniques

In this project I am creating it by using multiple analytical approach to make sure that it is technically feasible and secure with a great user experience.

- **Object-Oriented Analysis:** There are some attributes and interrelations, that has been used to implement a primary entities such as User, Product, and Order.
- **Data Pipeline Mapping:** Using Express Api to trace all the movements to the MongoDB database through frontend NEXTJs.
- **Use Case Documentation:** The system are recording all the actors interactions between actors and also outliers too.
- **Security Assessment:** Implemented the JWT authentication, encrypted the password hashing, and payment gateway to ensure user data security.
- **Performance Analysis:** While indexing the databases and optimizes the API response times to handle traffic and manage the large book inventories.

- **User Experience Optimization:** Trying to create an platform where everything can be easy to use with a high rate for both buyers and sellers by focusing and clarifying the information structure.

4. Progress Achieved:

4.1 Completed Tasks

The time duration of the report could see the upcoming successful of the BOOK_SHOP, and move to a fully integrated payment ecosystem.

❖ **Frontend Development:**

- Firstly, set up the NextJs, typescript and tailwind css
- Then use Redux toolkit store setup
- Use API service to connect with the Backend
- Design the UI of login, register along with the email authentication
- After that design the email verification page and the password reset page
- Now the header part designs with banner and feature sections, add footer components with the navigation menu for responsive bar
- Add Filter and pagination for listing the books
- Designing the book detail in page, and UI for shopping cart
- Design the wishlist & user profile page
- Used React hook for validation
- Installed package for the toast notifications
- Use the error handling along with the react route
- Import Next.js image component

❖ **Backend Development:**

- Firstly, setup the express.js server with TypeScript
- Then use the MongoDB with cluster and set 0.0.0.0 internet connection and configuration
- After that, set up the authentication system
- The important thing, that generates the JWT token, along with passport.js for google oAuth
- Added the tokens for email verification and password reset functionality
- Here is also used CRUD operations for products, cart management system, also wishlist functionality
- Having address management system with Order model and process the order.
- Trying to add payment method with SSLCommerz, bKash, GPay, Nagad
- Use cloundinary for image uploading

❖ **Payment Integration:**

- Integration the payment gateway of SSLCommerz mainly
- Have session creation, validation and verification
- Having the transaction status query, refund initiation and status tracking
- Check if payment success/failure/cancel with handling
- IPN (Instant Payment Notification) support

4.2 Results Obtained

The development levels produced a fully merged production with a ready prototype of this Book Shop. The preliminary results show that the chosen MERN stack with Next.js can effectively manage the difficulties of a student to student marketplace.

❖ **Functional Results (User-Facing):**

- **Multi-channel authentication:** Implementing this both Google and email authentication with validation and verification, reduces the hassle for the students specially.
- **End-to-End commerce Flow:** Real-time transaction processing was made possible by our successful integration of the wishlist and shopping cart logic with a live SSLCommerz payment gateway.
- **Experimental marketplace:** After the outcome, real buyers now can monitor the whole order process along with the progress. For now, we design a responsive dashboard that could be fit in both OS system.
- **Secure data:** All of the API are now can be protected by the authentication middleware, along with the guarantee that could use only authorized user data and financial record to access.

❖ **Technical Findings & Achievements:**

- **Structural Security:** After applying the typescript in the project, the runtime errors, syntax error, or any types of error are now solved easily along with enhancing the consistency without any distraction.
- **Data Performance:** The response time gets faster after the load testing for MogoDB design along with the Restful Api.
- **State & validation availability:** By using Redux along with the Zod, there is no error in software system. For that reason, error alert comes in handy during complex inputs and Zod, the schema of typescript doing well to management the validation library besides checks the information while running.
- **Optimized Asset Storage:** The cover will be effective after applying the high resolution, it should not causing any problem. By doing this, the server did not fall behind to the integration of a cloud based image storage system.

5. Challenges Faced:

S.No.	Issues and Challenges	Strategies or Plans
1	Payment gateway	For ensuring the erro handling in SSLcommerz with the sandbox environment.
2	State management issues	Using the Redux toolkit to manage the interplay between Next.js's server-side nature and Redux's client-side state management requirements
3	Image upload and storage	Cloudinary used for optimized image and video delivery, transformations, and uploading capabilities
4	Email configuration	Configured Nodemailer with SMTP and environment variables; created reusable email templates with retry logic.

5	Typescript type safety	Defined comprehensive interfaces for all models and enforced strict type checking via tsconfig.json.
6	CORS configuration Issues	CORS used to control the access for getting resources from different domains with the dynamic and credential support.
7	Password reset token	Implemented hashed, time-limited tokens stored in the database with strict expiration validation.
8	Order status	developed a state machine with built-in notifications for changes in status during transitions (such as Pending to Shipped).
9	Responsive design UI	Install the package of Tailwind CSS for the well design responsive features and adopt with mobile device also.
10	Authentication token	Used HTTP-only cookies for JWT storage to prevent XSS attacks; implemented token refresh mechanisms.
11	Database query optimization	Created Mongoose indexes for search fields and implemented Pagination to handle large book datasets.
12	Payment callback handling	For integrating the IPN system for the verification, have to develop custom success/fail/cancel endpoints
13.	Security Advisory: CVE-2025-66478 in NextJs/ReactJs	Vulnerability in Router security, so changed it to maintaining version with the patched version, and install this commands “ npm install next@15.5.9, npx fix-react2shell-next”

6. Next Steps:

The strategy below outlines the planned maintenance and upgrades to transform the online bookshop from its current state of operation into a strong, high-performing marketplace.

S.No.	Next Task	Estimated Completion Time (MM-YY)
1	Admin Dashboard	01-25
2	Advanced Search	01-25
3	Order Tracking	02-25
4	Reviews & Ratings	02-25
5	Seller Dashboard	02-25
6	Notification System	03-25
7	bKash / SSLcommerz Integration	03-25
8	Buyer-Seller Chat	03-25
9	Recommendation Engine	04-25
10	Analytics & Reporting	04-25

11	Mobile Application	05-25
12	Performance Optimization	05-25
13	Testing Suite	06-25
14	Security Audit	06-25
15	CI/CD Pipeline	07-25

7. Updated Timeline:

I am providing an updated timeline, highlighting progress and adjustments that I have made:

Tasks	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Backend & DB Design																		
Authen-tication																		
Product & Cart /Wishlist																		
Payment & Orders																		
Frontend Developme nt																		
Testing & Bug Fixes																		
Deployme nt																		

Estimated Work Period	
Actual Work Period	

8. Resources Utilized:

The development of the Online Book Shop relied on a modern full-stack ecosystem, integrating robust software frameworks with reliable third-party cloud services to ensure scalability and security. The following software stack was utilized to build, test, and document the platform:

Category	Tools & Technologies
Install environment	Node.js runtime, npm(package manager), VS Code editor, Git & Github
Frontend frameworks	Next.js, typescript, Tailwind css, Framer motion
State & form management	Redux Toolkit, React hook form, Radix UI
Backend frameworks	Express.js, Passport.js(auth), JWT(tokens), Multer(file handling)
Database	MongoDb, Mongoose
Payment & Services	SSLCommerz, bKash
API Testing	Postman

❖ Hardware Resources

The hardware was used to give a smooth facilities lifecycle development:

- **Development machine:** For development to get a well performed, we can use any platform like microsoft windows, Linux, or macOS.
- **Network structure:** Install a real-time API integration, package installations, and cloud synchronization, the internet should be flawless.
- **Storage:** The local disk used for source code storage and the image based resource have the cloudinary storage.

❖ Third-Party Services

External integrations were critical for extending the application's functionality:

- **Cloudinary:** It is used for optimizing the cloud based image management and hosting of book covers.
- **Payment Gateways:** We have to set a connection with **SSLCommerz** and **bKash** for securing the financial transactions and make it secure.
- **Communication:** The **Gmail SMTP** server will be used for the verification of email and automated email notifications.
- **Authentication:** **Google OAuth** is used for securing the social while login with capabilities.

9. Project Management and Financial Analysis:

The project needs to follow a structured framework for guaranteeing, that will be charge a good cost with a high quality delivery. For now, the main goal is to strike a balance between robust structure maintenance along with the fast development.

The investment will needed for both development and operations which is given below:

Category	Estimated Cost / Details
Development Cost	₹1,200,000 - ₹3,000,000 (Based on 400 - 500 hours at ₹3,000 - ₹6,000/hours)
Monthly Operating Costs	₹6,000 - ₹36,000 / month (Infrastructure & third-party services)
Transaction Fees	2% - 3% per transaction (SSLCommerz, bKash, GPay, Nagad)
Infrastructure	Includes hosting (Vercel/AWS), MongoDB Atlas, and Domain registration

❖ Project Management Approach

The project is using various types of strategies to maintain a well design and developed code in a cooperation level:

- **Agile development:** It is using an iterative process with regular tests for improving the development.
- **Quality check:** The demanding quality of code revisions and reviews should be checked at every stage.
- **Management Tools:** Git is usually used for checking the progress of task locally.
- **Documentation:** The user manual will be updated thrice a week on a regular basis in the entire life cycle.

❖ Budget Allocation

Resources are distributed to prioritize the main functionality and security of the marketplace:

- **Development (60%):** The primary focus will be on building main features.
- **Integration (20%):** The services which are from third parties, used for connecting like SSLCommerz and bKash.
- **Quality Assurance (10%):** Complete testing and bug fixing.
- **Documentation & Deployment (10%):** There will be final setup with hosting, and project documentation.

10.Future Considerations:

In this project, there will be a problem with the payment gateway because we do not have any merchant account, merchant account is really hard to get until the shop is eligible. The Book-Shop has a good potential with some problems, but also has its solution which is ensuring the long-term stability.

Core Technical & Operational Considerations:

Area	Issue	Problem solution
Scalability	Due to its growing user base, the database queries become slow	Have to implement the Redis caching and query improvement in practices.
Payment Reliability	A single third-party provider has high dependencies	Make sure, have an extra backup with alternative strategies.

Security	Have serious threats, data breaches, unauthorized access etc.	Should use the rate limiting, HTTPS, and strict validation.
Storage Costs	Need to increase the Cloudinary expenses with a huge image volume.	For increasing the size without losing it's resolution at a low cost, must apply image compression and CDN optimization.
Responsiveness	UI is hard to maintain with across different devices.	Make sure developers will be given the responsive design before from client or make it themselves, continuous cross-device testing.

❖ **Business & Performance Considerations**

- **Order fulfillment:** The physical deliveries is difficult to manage. For that, collaborating with delivery services which is very helpful nowadays, there are many delivery of API available like Pathao, speedfast, BDRex, eCourier, and our goal is to reduce this.
- **Legal Safety:** The system will implement a strict privacy policies in terms of service (GDPR) which is supported by regular legal reviews and it will address the data privacy issues.
- **Performance Efficiency:** In the future, personal support service system will be added to offer offline support to know the problem. And also fix the loading system which is delayed for some technical issues would be solved.
- **User support facilities:** Customer questions will be answered by a AI assistant Bot. Live chat and ticketing systems are being considered for the next phase.
- **Feature Roadmap:** We will maintain a continuous feedback loop with users in order to maintain the plan feature enhancements and ensure that the platform grows according to market demands.

11.Conclusion:

The Book Shop project are presenting to you a comprehensive e-commerce web designed with the availability of both buying and selling of books in Bangladesh. Through the implementation of modern web, secure payment processing, and intuitive user interfaces, this service are able to fill up the expanding gap for easy-to-use book trading.

While developed a full-stack e-commerce with some exceptional features like secure authentication system with multiple login options, SSLCommerz payment integration, responsive UI design, robust database schema, and scalable API architecture design. This happened because all of this was built with typescript for type safety, Next.js for modern React-based server-side rendering, Express.js for RESTful APIs, and MongoDB for data management/database, along with the error handling and validation.

Future Expansion:

In the next few future ,we will have a strong foundation with growth, along with the search in advanced with recommendations, dashboards of seller/admin, analytics or reporting and enhanced communication tools.

This project elaborates it's proficiency in this MERN-stack development, API connectivity for database, and tell how to implement payment gateway in modern web development. The architecture is allowing easy maintenance and future enhancements, making it a sustainable and scalable solution for the book trading marketplace.

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Appendix

Appendix A: Project Structure

FINAL YEAR DESIGN PROJECT

PHASE-I PROGRESS REPORT

This report, in the form of a template, has been specifically designed for BSc. students working on their Final Year Design Project (FYDP) at Computer Science and Engineering Department, Daffodil International University (DIU).

Every group of students is required to do the following:

1. Complete all the sections of this template
2. Get it certified by the assigned supervisor before one week of Phase-I evaluation presentations
3. Submit 01 photocopy to each of the following, on or before the day of Phase-I presentations:
 - a. Supervisor
 - b. Internal Evaluator
4. Submit original copy to FYDP committee on the day of Phase-I presentations.

Note:

1. Use English
2. There should be NO grammatical or spelling mistakes
3. Submission after due date will not be accepted
4. For more information, contact your Supervisor

Template prepared by:	Template approved by:
FYDP Committee Dept. of CSE, DIU	Dr. Sheak Rashed Haider Noori Professor and Head, Dept. of CSE, DIU

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